B.Tech CSE, II Year I Sem, BH25 curriculum, BVRIT Hyderabad

Restaurant Ordering and Billing System

Java Lab Project, Object Oriented Programming Through Java Lab (CS306PC) .

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Problem statement and objective

- **Problem: Small restaurants handle menus, orders and bills manually. This causes calculation errors, slow billing and no sales records.**
- **Objective: Build a Java application that manages a menu, takes orders, applies discounts, adds 6% tax and generates bills.**
- **Secondary objective: Apply the lab's OOP, collections, streams, exception handling and Swing concepts in one project.**

Features

- Two restaurants - KFC (7 items) and Dominos (6 items), each with its own menu
- Menu management - Add new items; duplicate names/ids and prices ≤ 0 are rejected
- Ordering - Cart with quantities; order ids start at 1001
- Discounts - None, Flat Rs.50 off, 10% off
- Billing - Subtotal → discount → 6% tax → total
- Bill saving - Console version saves bills/bill_<id>.txt
- Reports - Veg items, price filter, top sellers, uppercase names
- Search - Item lookup by name (not case-sensitive)
- Split bill - GUI only; amount per person, divide-by-zero handled
- Two interfaces - Console menu and Swing GUI sharing the same logic

Technology

- Language: Java (Java 8+ features used: lambdas, streams). Tested on JDK 21.
- GUI: Java Swing and AWT (JFrame, JTable, JList, JComboBox, GridLayout, BorderLayout)
- Tools: VS Code / Eclipse, command line javac and java
- No external libraries or database

5. Architecture: classes

- Restaurant (interface): Defines the operations every restaurant must provide: printName, addItem, getMenu, placeOrder, generateBill, billText, findByName and getSalesCount. It also holds the constant TAX_RATE = 0.06.
- KFC (class): Implements Restaurant. It stores the menu in an Item[] array, keeps orders in a map, and does the validation and bill calculation.
- Dominos (class): Extends KFC and supplies its own name and menu, reusing all the other logic.
- Item (bean): Holds an item's id, name, price, category and veg/non-veg flag.
- Order (bean): Holds the order id, the item-to-quantity lines and the chosen discount.
- Discount (abstract class): Declares apply(amount) and label() for every discount type.
- NoDiscount, FlatDiscount, PercentDiscount (subclasses): The three concrete discounts: none, flat Rs.50 off and a percentage off.
- InvalidOrderException (custom exception): Raised for an empty cart, unknown item or order, bad quantity, invalid price or duplicate item.
- Reports (utility class): Stream-based reports (veg items, price filter, top sellers, uppercase names) and the next item id.
- ConsoleApp (UI): The switch-based console menu.
- RestaurantGUI (UI): The Swing window with the menu table, keypad, cart and bill output.
- Main (entry point): Asks for console or GUI and starts the chosen front end.

Java concepts demonstrated

Interface, inheritance, abstract class, method overriding, polymorphism and late binding (Restaurant r = new KFC() or new Dominos()), encapsulation (private fields with getters), custom checked exception, HashMap, arrays, LinkedHashMap, Stream API (filter, map, sorted, limit), lambdas, switch...case, file I/O (Files.write), Swing event handling, multi-catch.

7. Lab programs covered (11)

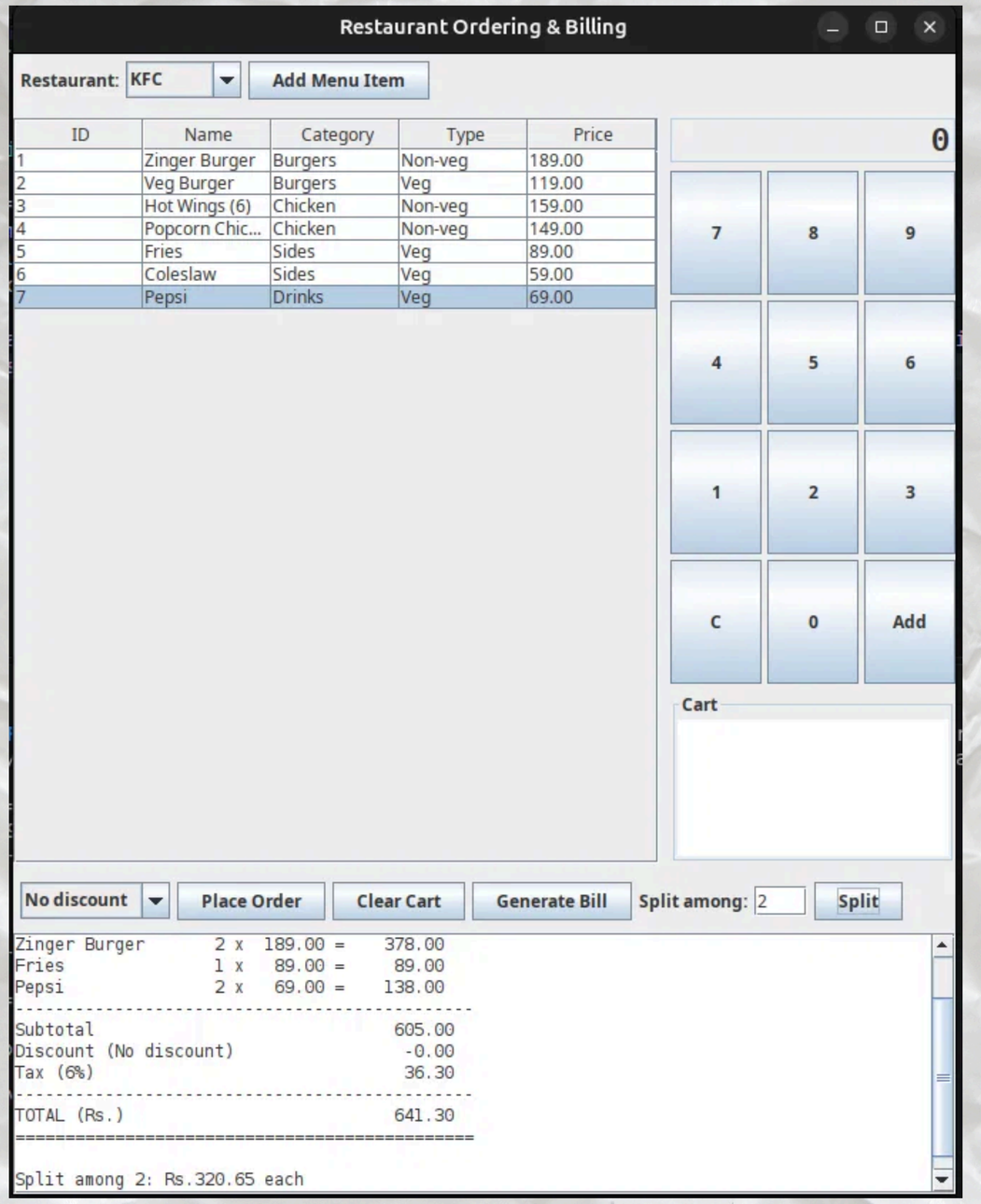
Prog	Concept	Where
3	switch...case	Console menu
12	Interface, late binding, Item[]	`Restaurant`, `KFC`
13	Abstract class	`Discount`
14	Exception handling	`InvalidOrderException`
18	Bean classes	`Item`, `Order`
19	HashMap lookup	`findByName`
22	HashMap frequency count	Sales count
23, 24	Streams	`Reports`
25, 26	Swing GUI	`RestaurantGUI`

Working flow

View menu → Build cart (item id + quantity) → Choose discount → Place order (validated, order id issued) → Generate bill → Save bill (console).

Bill formula: Total = (Subtotal − Discount) + 6% × (Subtotal − Discount), which equals (Subtotal − Discount) × 1.06.

GUI layout

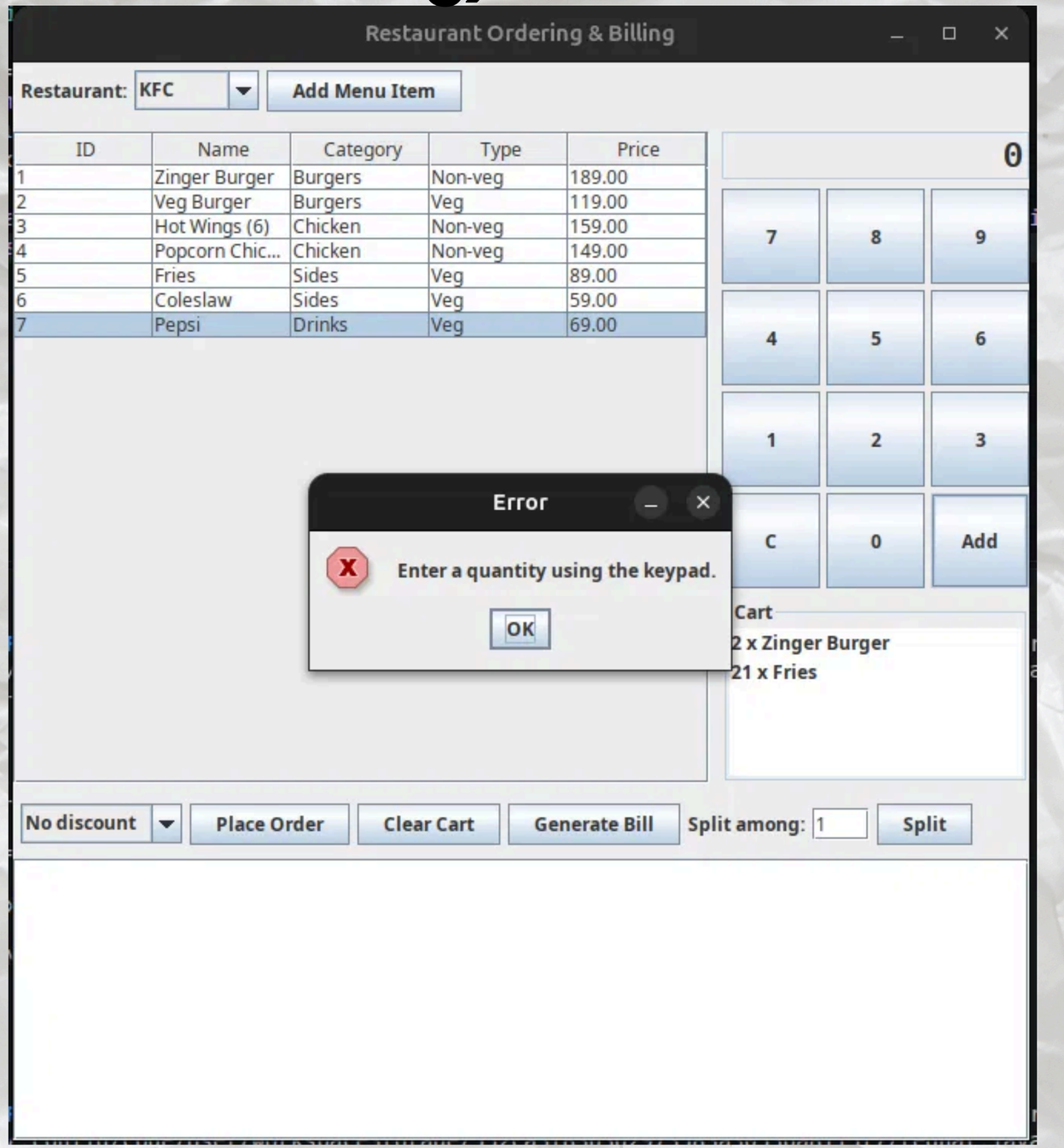

Restaurant Ordering & Billing
Restaurant: KFC
Add Menu Item
ID Name Category Type Price
1 Zinger Burger Burgers Non-veg 189.00
2 Veg Burger Burgers Veg 119.00
3 Hot Wings (6) Chicken Non-veg 159.00
4 Popcorn Chic... Chicken Non-veg 149.00
5 Fries Sides Veg 89.00
6 Coleslaw Sides Veg 59.00
7 Pepsi Drinks Veg 69.00
0
7 8 9
4 5 6
1 2 3
C 0 Add
Cart
No discount
Place Order
Clear Cart
Generate Bill
Split among: 2
Split
Zinger Burger 2 x 189.00 = 378.00
Fries 1 x 89.00 = 89.00
Pepsi 2 x 69.00 = 138.00
Subtotal 605.00
Discount (No discount) -0.00
Tax (6%) 36.30
TOTAL (Rs.) 641.30
Split among 2: Rs.320.65 each

- **Console menu**

- 1 View Menu
- 2 Add Item
- 3 Place Order
- 4 Generate Bill
- 5 Reports
- 6 Search
- 7 Change Restaurant ·
- 0 Exit

```
cse@admin-cse:~/Downloads/RestaurantBilling$ 1
1: command not found
cse@admin-cse:~/Downloads/RestaurantBilling$  cd /home/cse/Downloads/RestaurantBilling ; /usr/bin/env /usr/lib/jvm/j
tionMessages -cp /home/cse/.config/Code/User/workspaceStorage/552ca3f83b3d2973583a30536a8115f7/redhat.java/jdt_ws/Re
1 = Console, 2 = GUI : 1
Restaurant (1 = KFC, 2 = Dominos): 1
Restaurant: KFC

========= RESTAURANT SYSTEM =========
Restaurant: KFC
1.View Menu   2.Add Item   3.Place Order   4.Generate Bill
5.Reports     6.Search     7.Change Restaurant   0.Exit
Choice: 1
ID   Name                  Category   Type     Price
1    Zinger Burger         Burgers    Non-veg  Rs.189.00
2    Veg Burger            Burgers    Veg      Rs.119.00
3    Hot Wings (6)         Chicken    Non-veg  Rs.159.00
4    Popcorn Chicken       Chicken    Non-veg  Rs.149.00
5    Fries                 Sides      Veg      Rs.89.00
6    Coleslaw              Sides      Veg      Rs.59.00
7    Pepsi                 Drinks     Veg      Rs.69.00

========= RESTAURANT SYSTEM =========
Restaurant: KFC
1.View Menu   2.Add Item   3.Place Order   4.Generate Bill
5.Reports     6.Search     7.Change Restaurant   0.Exit
Choice: 3
ID   Name                  Category   Type     Price
1    Zinger Burger         Burgers    Non-veg  Rs.189.00
2    Veg Burger            Burgers    Veg      Rs.119.00
3    Hot Wings (6)         Chicken    Non-veg  Rs.159.00
4    Popcorn Chicken       Chicken    Non-veg  Rs.149.00
5    Fries                 Sides      Veg      Rs.89.00
6    Coleslaw              Sides      Veg      Rs.59.00
7    Pepsi                 Drinks     Veg      Rs.69.00
Item id (0 to finish): 5
Quantity: 2
Item id (0 to finish): 0
Discount (1 = None, 2 = Flat Rs.50, 3 = 10%): 3
Order placed. Order id = 1001

========= RESTAURANT SYSTEM =========
Restaurant: KFC
1.View Menu   2.Add Item   3.Place Order   4.Generate Bill
5.Reports     6.Search     7.Change Restaurant   0.Exit
Choice: 7
Restaurant (1 = KFC, 2 = Dominos): 2
Restaurant: Domino's

========= RESTAURANT SYSTEM =========
Restaurant: Domino's
1.View Menu   2.Add Item   3.Place Order   4.Generate Bill
5.Reports     6.Search     7.Change Restaurant   0.Exit
Choice: 1
```

Sample output (verified in testing)



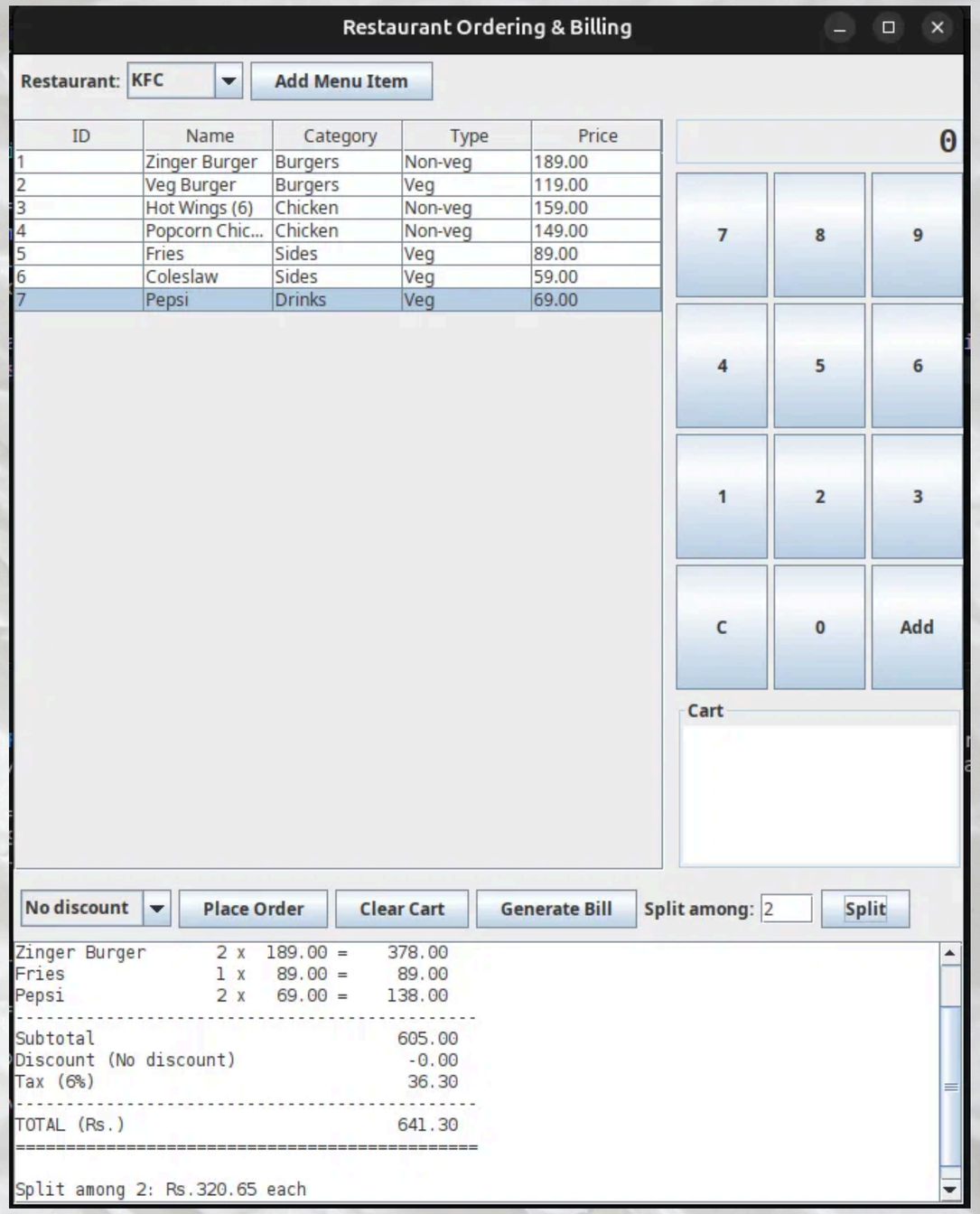
Sample output (verified in testing)

bills > bill_1001.txt

```
==========================================
  Domino's  |  Order #1001
==========================================
Margherita Pizza    6 x  199.00 =   1194.00
------------------------------------------
Subtotal                            1194.00
Discount (Flat Rs.50 off)            -50.00
Tax (6%)                              68.64
------------------------------------------
TOTAL (Rs.)                         1212.64
==========================================
```

Problems 1 | Output | Debug Console | Terminal | Ports

```
cse@admin-cse:~/Downloads/RestaurantBilling$ cd /home/cse/Downloads/RestaurantBilling ; /usr/bin/env /usr/lib/jvm/j
tionMessages -cp /home/cse/.config/Code/User/workspaceStorage/552ca3f83b3d2973583a30536a8115f7/redhat.java/jdt_ws/Re
Choice: 4
Order id (0 = last order): 0
==========================================
  Domino's  |  Order #1001
==========================================
Margherita Pizza    6 x  199.00 =   1194.00
------------------------------------------
Subtotal                            1194.00
Discount (Flat Rs.50 off)            -50.00
Tax (6%)                              68.64
------------------------------------------
TOTAL (Rs.)                         1212.64
==========================================

Saved to bills/bill_1001.txt

========= RESTAURANT SYSTEM =========
Restaurant: Domino's
1.View Menu   2.Add Item   3.Place Order   4.Generate Bill
5.Reports     6.Search     7.Change Restaurant   0.Exit
Choice: 0
Thank you!
cse@admin-cse:~/Downloads/RestaurantBilling$
```



Limitations

- Data is kept in memory only; menu additions and orders are lost when the program closes.
- The GUI does not save bills, and has no Reports or Search screens.
- No login, no payment handling, no database.

Future scope

Database storage (JDBC), coupons and multiple tax slabs, table-wise billing, user login (admin/cashier), printable PDF bills.

Conclusion

A working Restaurant Ordering and Billing System that applies 11 lab programs in one application, with console and GUI versions, validated input and accurate billing